

BIOINFORMATICS MINI-PROJECT 3

Project 3 — Protein Sequence Similarity Analysis

Hen Egg-White Lysozyme Using BLASTP and COBALT

Student End Report

Beginner-level research-profile project

1. Abstract

This project introduced protein sequence analysis using **NCBI Protein BLAST (BLASTP)**. A 129-residue lysozyme sequence was searched against protein databases. The strongest visible result was hen egg-white lysozyme from *Gallus gallus* with 129/129 identical residues (100%), no gaps, and an E-value of 4e-89. Closely related avian lysozymes also showed very high similarity. COBALT was opened as a multiple-sequence-alignment step, but its recorded graphical overview was “Not available,” so no completed COBALT alignment interpretation is claimed.

2. Objective

Which known protein sequences are most similar to the lysozyme sequence being studied?

3. BLASTP in Simple Terms

BLASTP is like a search engine for protein sequences. A protein sequence is submitted and compared with sequences in a database. The results help identify proteins, find homologues, and examine sequence conservation.

4. Query Sequence

The query began **KVFGRCELAAAMKRHGLDNRYRGYSLGNWVCAAKFESNFNTQATNRNTDGSTDY...** and represented the 129-residue lysozyme sequence used in the earlier structural projects.

5. Main BLAST Results

Hit	Identity	Positives	Gaps	E-value	Score
1LSG_A — HEN EGG WHITE LYSOZYME [Gallus gallus]	129/129 (100%)	129/129 (100%)	0/129	4e-89	266 bits (679)
XP_015711651.2 — lysozyme C [Coturnix japonica]	123/129 (95%)	126/129 (97%)	0/129	2e-84	254 bits (650)
ACL81751.1 — lysozyme [Bambusicola thoracicus]	123/129 (95%)	124/129 (96%)	0/129	2e-84	254 bits (649)

6. How to Read the Results

Identity means the exact same amino acid occurs at an aligned position. **Positives** include identical residues plus substitutions considered chemically similar. **Gaps** represent inserted alignment positions. **E-value** describes the expected number of matches of similar score that could occur by chance; smaller values indicate stronger statistical significance. **Bit score** summarizes alignment strength.

7. Top Hit Interpretation

The top visible result, **1LSG_A — HEN EGG WHITE LYSOZYME [Gallus gallus]**, aligned the full 129-residue query with 129/129 identities, 100% positives and no gaps. The recorded alignment placed query residues 1–129 against subject residues 2–130. In simple terms, the query matched the reported hen egg-white lysozyme sequence exactly across all query residues.

8. Closely Related Avian Hits

The next visible hits included lysozyme from *Coturnix japonica* and *Bambusicola thoracicus*. Each showed 123/129 identities (95%) with no gaps. This illustrates strong sequence conservation among related bird lysozymes.

9. COBALT / Multiple Sequence Alignment

COBALT was opened after BLAST. The returned sequence set included the query and many lysozyme-related sequences. However, the recorded COBALT page displayed **Graphical Overview: Not available**. Therefore, this report documents the attempted MSA step honestly and does not present a completed alignment analysis.

10. Biological Interpretation

The BLAST results provide strong sequence-level support that the query is hen egg-white lysozyme. The exact *Gallus gallus* match is consistent with Projects 1 and 2, while the highly similar avian hits introduce the idea of sequence conservation across species.

11. What I Learned

I learned that BLAST results are more than protein names: identity, positives, gaps, E-value and score describe different aspects of a sequence match. I also learned to distinguish a completed analysis from a tool step whose output was not available.

12. Limitations and Future Work

No full phylogenetic analysis, domain analysis or completed COBALT interpretation was performed. Future work should complete the multiple sequence alignment, identify conserved and variable residues, build a simple phylogenetic tree, and map sequence conservation onto the 3D lysozyme structure.

13. Conclusion

BLASTP identified a 100% identical 129-residue hen egg-white lysozyme match and very similar avian lysozyme sequences. The project therefore provides a clear sequence-level confirmation of protein identity and a first introduction to comparative sequence analysis.

14. References

NCBI BLAST: <https://blast.ncbi.nlm.nih.gov/>

NCBI BLAST FAQ: https://blast.ncbi.nlm.nih.gov/Blast.cgi?+CMD=Web&DOC;_TYPE=FAQ&PAGE;_TYPE=BlastDocs

RCSB PDB 1FDL: <https://www.rcsb.org/structure/1FDL>

Appendix — Project Screenshots

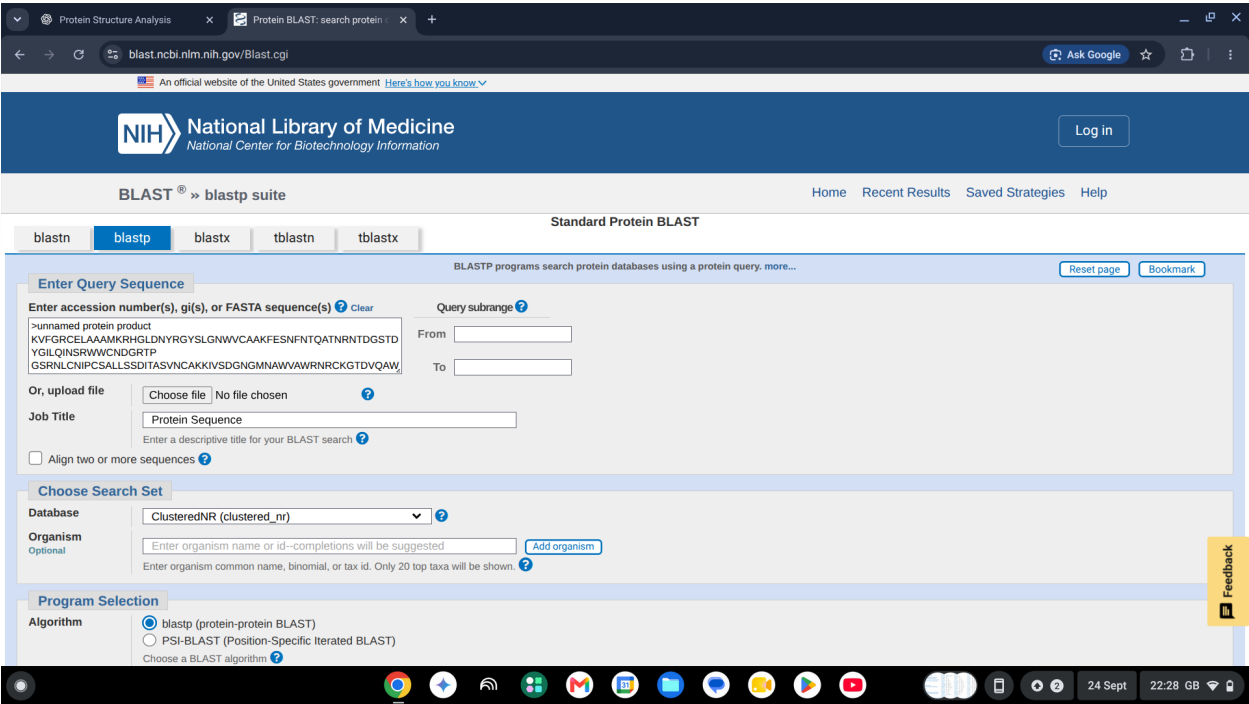


Figure A1. NCBI Protein BLAST input page showing the query.

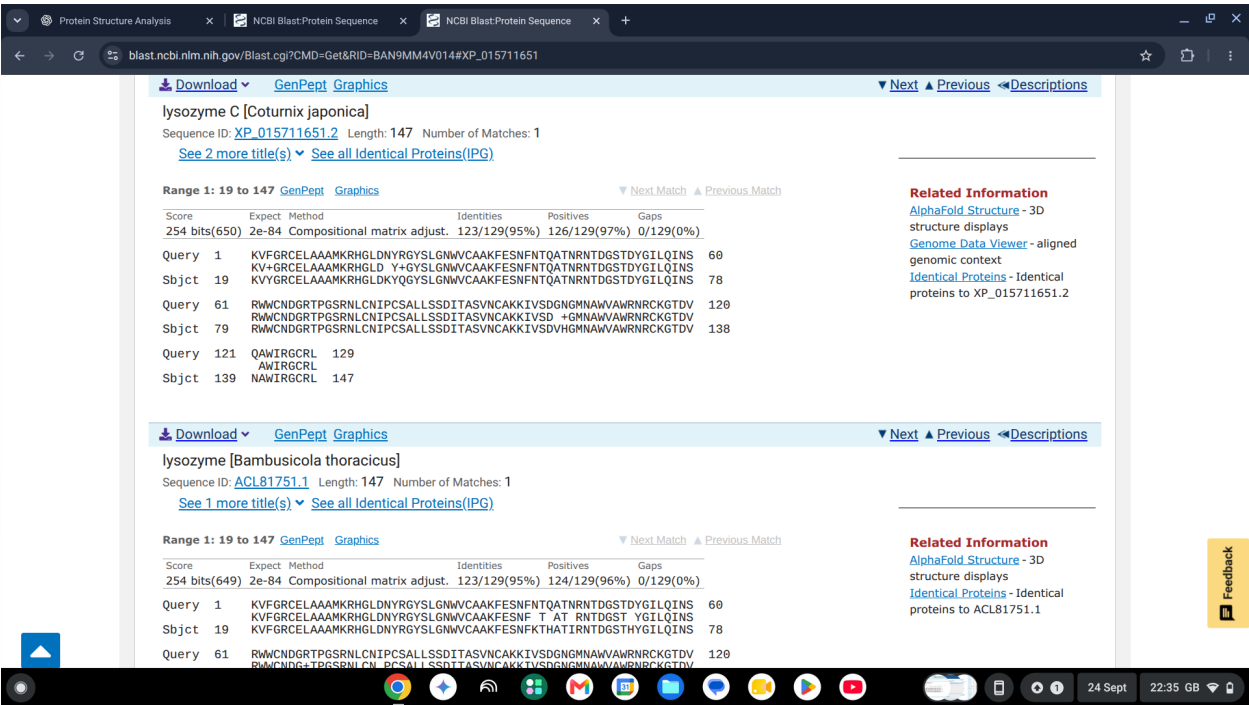


Figure A2. BLAST pairwise/result view showing the 100% identity top match.

The screenshot shows the NCBI BLAST results page for a protein sequence analysis. The browser tabs include "Protein Structure Analysis", "NCBI Blast-Protein Sequence", and "COBALT - Cobalt RID (seq)". The URL is "blast.ncbi.nlm.nih.gov/Blast.cgi". The page header features the NIH logo and "National Library of Medicine National Center for Biotechnology Information". The main heading is "BLAST® » blastp suite » results for RID-BAN9MM4V014". Below this, there are links for "Home", "Recent Results", "Saved Strategies", and "Help".

The "Job Title" is "Protein Sequence". The "RID" is "BAN9MM4V014", with a note "Search expires on 09-26 00:52 am" and a "Download All" link. The "Program" is "BLASTP", and the "Database" is "nr". The "Query ID" is "IcIQuery_6305759". The "Description" is "unnamed protein product". The "Molecule type" is "amino acid". The "Query Length" is "129". Other reports include "Distance tree of results", "Multiple alignment", and "MSA viewer".

The "Filter Results" section includes a "Filter" button and a "Reset" button. The "Organism" filter is set to "only top 20 will appear". The "Percent Identity" filter is set to "to". The "E value" filter is set to "to". The "Query Coverage" filter is set to "to".

The "Sequences producing significant alignments" section shows a table with columns for "Accession", "Description", and "Links". The table lists various protein sequences, including "Chain A, HEN EGG WHITE LYSOZYME [Gallus gallus]", "Chain A, LYSOZYME C [Gallus gallus]", and "Chain A, HEN EGG WHITE LYSOZYME [Gallus gallus]".

The bottom of the page shows a taskbar with various application icons and system information: "24 Sept", "22:23 GB", and a battery icon.

Figure A3. BLAST results page used to inspect ranked matches.

The screenshot shows the NCBI COBALT sequence list page. The browser tabs include "Protein Structure Analysis", "NCBI Blast-Protein Sequence", and "COBALT - Cobalt RID (seq)". The URL is "ncbi.nlm.nih.gov/tools/cobalt/cobalt.cgi". The page displays a table with columns for "Accession", "Description", and "Links". The table lists various protein sequences, including "Chain A, HEN EGG WHITE LYSOZYME [Gallus gallus]", "Chain A, LYSOZYME C [Gallus gallus]", and "Chain A, HEN EGG WHITE LYSOZYME [Gallus gallus]".

A "Screen capture" window is open in the bottom right corner, showing a screenshot of the BLAST results page. The window includes a "Copied to clipboard" button and a "Launcher + V" button.

The bottom of the page shows a taskbar with various application icons and system information: "24 Sept", "22:39 GB", and a battery icon.

Figure A4. COBALT sequence list returned for the multiple-alignment step.

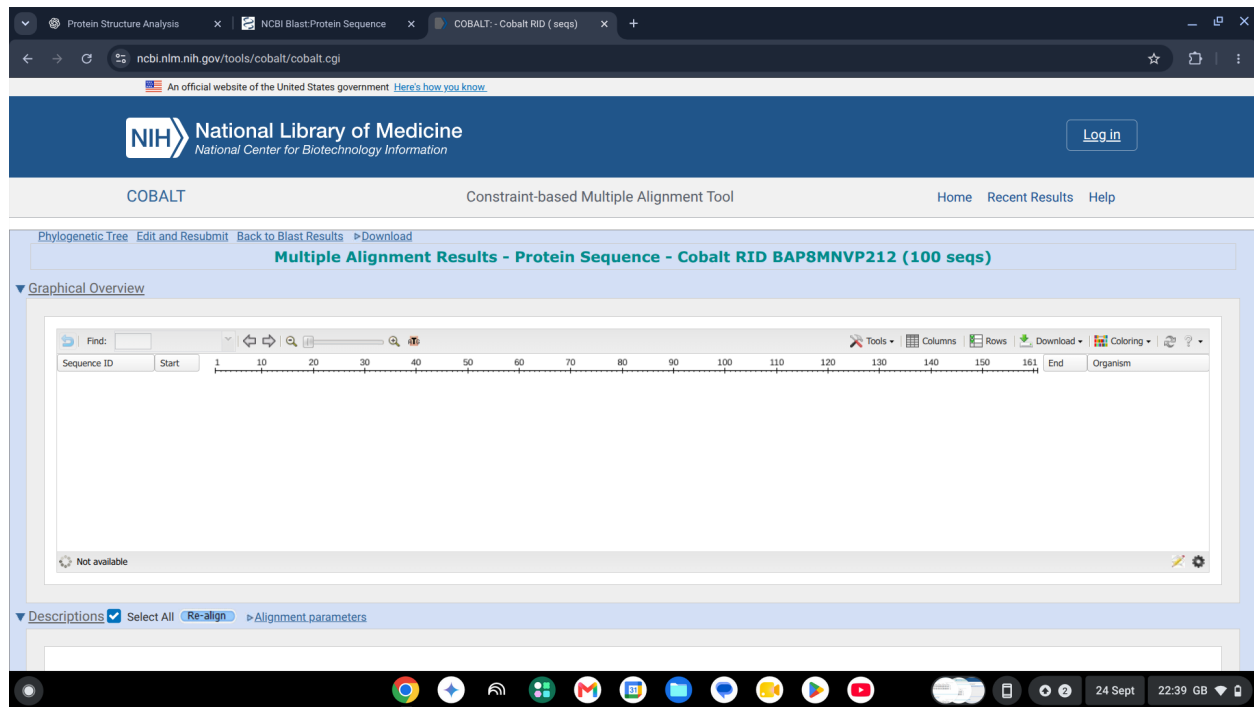


Figure A5. COBALT page showing the graphical overview as 'Not available'.